

THE HORIZON PROGRAM

OBLIVION I

Official Mission Dossier
Version 7.0 — Updated April 22, 2026

"The horizon always has something beyond it. But if we fall short, there is nothing after."

Rocket	Oblivion XXX (engine count TBD)
Spacecraft	ARES-1 — Advanced Rapid Exploration Spacecraft
Objective	Crewed Mars Orbital Insertion — First humans to orbit Mars
Crew	8 (selection pending)
Duration	~500 to 580 days
ARES-1 Design	Vertical integrated vehicle — 150 m x 15 m
Assembly	7 modules — orbital assembly — NTP ignites from orbit only
Refueling	Two-point: Earth staging orbit + Low Mars Orbit depot (x2)
Test Flight	Oblivion I-T — Uncrewed full-fidelity precursor
Total Budget	\$1,000,000,000,000 (One Trillion USD)
Coalition	USA / China / Russia / Saudi Arabia / Japan / India / UAE / Canada
Design Start	April 10 ARES-1 design finalized April 22

Foreword

This document is the complete, official, and binding mission dossier for Oblivion I and its preceding test flight, Oblivion I-T. Version 7.0 supersedes all previous versions. The primary changes from Version 6.0 are the finalization of the ARES-1 spacecraft design as a vertical integrated vehicle 150 metres tall and 15 metres wide, the confirmation of the seven-module orbital assembly architecture, and the formal adoption of the integrated NTP engine block architecture replacing the previously described linear truss separation design. All other program parameters remain as established in Version 6.0.

This dossier is the single source of truth for the entire program. When there is ambiguity, come back to this document. When there is disagreement, come back to this document. Everything flows from what is written here.

Part One — The Horizon Program

The Horizon Program is the overarching initiative under which all missions, vehicles, and activities in this dossier exist. It is humanity's first structured, funded, and internationally ratified program for deep interplanetary human exploration. Its name was chosen deliberately. A horizon is something you perpetually push toward — every time you reach one, a new one appears beyond it. Oblivion I is the first horizon. What lies beyond it will be defined by what it proves.

Part Two — The Coalition

The Horizon Program is funded and governed by an eight-nation international coalition — the most geopolitically diverse human spaceflight partnership ever assembled. It spans traditional geopolitical adversaries who have agreed, for the first time in history, to point their collective resources at a shared destination rather than at each other.

Nation	Est. Contribution (Both Missions)	Primary Role
United States	\$400–450 billion	Mission lead, NASA, Deep Space Network
China	\$150–200 billion	Manufacturing, CNSA Mars heritage
Russia	\$50–75 billion	Long-duration spaceflight expertise, life support
Saudi Arabia	\$100–125 billion	Sovereign Wealth Fund (PIF) financing
Japan	\$75–100 billion	Precision engineering, avionics lead
India	\$50–75 billion	Systems engineering, software talent pool
UAE	\$25–50 billion	Mars orbital mission heritage (2021)
Canada	\$25–50 billion	Robotic orbital assembly systems lead

Part Three — The Budget

Mission	Allocation
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Oblivion I-T — Uncrewed Test Flight	\$450,000,000,000
Oblivion I — Crewed Mission	\$550,000,000,000
TOTAL	\$1,000,000,000,000

Both allocations are fully independent. Neither borrows from the other. Both are subject to fluctuation only by unanimous coalition agreement.

Part Four — The Two-Mission Architecture

The coalition authorized two full missions — one uncrewed test flight funded at \$450 billion and one crewed mission funded at \$550 billion — each built to identical specification and each flying the complete mission profile. The uncrewed test flight exists to find every failure mode that cannot be discovered on the ground before human lives are committed. A test flight failure does not trigger the zero-funding clause. The crewed mission allocation remains fully intact regardless of the test flight outcome. Option C — abandoning the test flight and committing the full \$1 trillion to a single crewed attempt — was rejected. Boldness without a safety architecture is not courage.

Part Five — Oblivion I-T (Test Flight)

Designation	Oblivion I-T
Type	Uncrewed Full-Fidelity Deep Space Test Flight
Trajectory	Complete Mars Orbital Insertion — identical to crewed mission
Duration	~500 to 580 days
Budget	\$450,000,000,000
Crew	None

Oblivion I-T flies the complete mission profile autonomously — orbital assembly at Earth, Earth orbit refueling, trans-Mars injection burn, Mars orbital insertion, LMO depot rendezvous and refueling, Mars orbital operations, trans-Earth injection burn, return transit, capsule separation, skip reentry, and splashdown. Every system operates exactly as it will during the crewed mission.

Part Six — Oblivion I (The Crewed Mission)

Designation	Oblivion I
Type	Crewed Deep Space Mars Orbital Insertion Mission
Trajectory	Trans-Mars Injection — Mars Orbital Insertion — Trans-Earth Injection
Mars Orbit	Low Mars Orbit — duration TBD pending design phase
Duration	~500 to 580 days total
Budget	\$550,000,000,000
Crew	8 (selection pending)
Delta-V	~5,900 to 6,200 m/s (NTP ignites from orbit only)

This is the mission. Eight human beings will board ARES-1 in Earth orbit, commit to a trajectory that takes them further from home than any human being has ever traveled, enter orbit around Mars — the first humans to orbit another planet in history — conduct orbital science operations, refuel at the pre-positioned Low Mars Orbit depot, and return to Earth alive.

Maneuver	Delta-V
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Trans-Mars Injection burn from Earth orbit	~3,600 m/s
Mars Orbital Insertion burn	~900 m/s
Mars orbit operations and corrections	~200 m/s
Trans-Earth Injection burn (from LMO depot)	~1,100 m/s
Earth return trajectory corrections	~100 m/s
TOTAL	~5,900 to 6,200 m/s

Part Seven — ARES-1 Spacecraft (Finalized Design)

ARES-1 is a vertical integrated spacecraft — the most ambitious crewed vehicle ever designed. Unlike the original linear truss concept, the finalized design is a compact tower 150 metres tall and 15 metres wide, with all systems integrated into a single coherent structure. The NTP engine block is at the base, the service module occupies the central 100 metres, and the crew sections occupy the upper 50 metres. This design was finalized on April 22, 2026.

Parameter	Value
Total Length	150 metres
Total Width	15 metres
Crew Capsule	20 metres — dome-shaped — 8 seats — headlights — RCS thrusters
Living Room / Habitat	30 metres — large windows — solar panels — RCS thrusters
Service Module	100 metres — 4 segments — service hatch — power/avionics panel
NTP Engine Block	Integrated side and bottom — 3 main engines — side engines
Solar Arrays	Deployable — 200 m ² — 60 kW at 1 AU — 26 kW at Mars
RCS Thrusters	All sides — pitch, yaw, roll authority throughout vehicle
Total Mass (fueled)	~250 to 300 metric tonnes in orbit
NTP Specific Impulse	~900 seconds — ignites from orbit only
Mission Delta-V	~5,900 to 6,200 m/s
Radiation Shielding	Tri-layer: hull + water wall + active HTS magnetic deflection
Communications	Deep Space Optical Comm — 200 Mbps downlink
Target Radiation Dose	Below 0.6 Sievert total per crew member
Earth Return Speed	12 to 14 km/s — skip reentry

Seven-Module Orbital Assembly Architecture

Module	Length	Dry Mass	Function
M1 — Crew Capsule	20 m	~15 t	Human-rated launch — crew boards here
M2 — Habitat / Living Room	30 m	~25 t	Pressurised crew living and work space

M3 — Service Seg A (top)	28 m	~30 t	Life support, avionics, science equipment
M4 — Service Seg B (mid)	28 m	~35 t	Power systems, water storage, comms
M5 — Service Seg C (bottom)	28 m	~40 t	LH2 propellant tanks — radiation buffer
M6 — Service Seg D (base)	16 m	~20 t	Structural interface, feed lines, RCS
M7 — NTP Engine Block	~12 m	~30 t	Nuclear reactor, 3 main + side engines
TOTAL DRY	~162 m	~195 t	+ ~80-100 t propellant via tanker missions

Part Eight — Complete Mission Flight Plan

Phase 1 — Orbital Assembly

Oblivion XXX flies 7+ missions delivering ARES-1 modules to Earth staging orbit at 400-600 km. Canada's robotic systems assemble each module. Ground teams verify every connection before the next module launches. Vehicle is structurally complete but carries no propellant.

Phase 2 — Earth Orbital Refueling

Oblivion XXX flies dedicated tanker missions delivering liquid hydrogen to the orbital depot. Once the full outbound propellant load is confirmed the depot transfers propellant to ARES-1's NTP tanks. Crew launches separately and docks. Final systems checks complete. Mission control gives go.

Phase 3 — Trans-Mars Injection

ARES-1's NTP engine ignites from orbit for the trans-Mars injection burn — the point of no return. The burn lasts 20 to 40 minutes. Engine cutoff. ARES-1 is on a heliocentric transfer trajectory. There is no abort back to Earth orbit.

Phase 4 — Outbound Transit (~200-260 days)

Eight crew members maintain continuous watch rotations on alternating cycles. All five systems monitored continuously. Mandatory two-hour daily exercise regime. Deep space science observations. Communication delay grows to 20 minutes at maximum distance.

Phase 5 — LMO Depot Rendezvous

ARES-1 begins autonomous rendezvous with the pre-positioned Mars depot 10-14 days before orbital insertion. Earth return propellant transfers from depot to ARES-1 tanks. Two depots pre-positioned — if primary fails, secondary covers. Both remain as permanent Mars infrastructure.

Phase 6 — Mars Orbital Insertion

ARES-1 fires NTP engine at ~900 m/s delta-V to decelerate into Low Mars Orbit. First humans to orbit another planet in history. Mars fills the observation windows.

Phase 7 — Mars Orbital Operations

Full science program from Low Mars Orbit. High-resolution surface imaging, atmospheric probe deployments, magnetosphere characterisation, fourteen CubeSat orbiters deployed in two salvos. Duration TBD pending design phase consumables analysis.

Phase 8 — Trans-Earth Injection

NTP engine fires for Earth return burn. Propellant loaded at LMO depot — did not travel from Earth. Engine cutoff. Mars recedes. The crew takes their last look at the planet they orbited.

Phase 9 — Return Transit (~200-260 days)

Return phase historically when crew psychological stress peaks in analog missions. Flight Surgeon's most demanding phase. Watch schedules, exercise, private quarters, and communications link home are all active life support systems.

Phase 10 — Capsule Separation, Reentry, Splashdown

Three days before Earth arrival crew transfers to capsule. Commander initiates separation. Main ARES-1 vehicle continues on trajectory. Biconic capsule performs skip reentry at 12-14 km/s. Drogue parachutes at 8,000 m. Three mains at 3,000 m. Splashdown. Eight crew members home.

Phase	Duration	Key Event
1 — Orbital Assembly	Weeks to months	ARES-1 assembled from 7 modules in Earth staging orbit
2 — Earth Refueling	Days	Full outbound propellant transferred from depot
3 — Trans-Mars Injection	20-40 min burn	Point of no return — NTP ignites from orbit
4 — Outbound Transit	~200-260 days	Deep space transit — continuous watch rotations
5 — LMO Depot Rendezvous	~3-5 days	Earth return propellant loaded at Mars
6 — Mars Orbital Insertion	20-40 min burn	First humans to orbit another planet
7 — Mars Orbital Operations	TBD design phase	Full science program — CubeSats — probes
8 — Trans-Earth Injection	20-40 min burn	Departure from Mars orbit
9 — Return Transit	~200-260 days	Return journey — capsule preparation
10 — Reentry & Splashdown	~30 minutes	Skip reentry at 12-14 km/s — crew recovery
TOTAL	~500-580 days	All eight crew members home alive

Part Nine — The Consequence of Failure

WARNING — READ THIS SECTION IN FULL BEFORE PROCEEDING WITH ANY DESIGN WORK.

If Oblivion I — the crewed mission — fails for any reason under any circumstances, the Horizon Program receives zero further funding and is immediately and permanently dissolved. There is no appeal. There is no restructuring. Eight sovereign nations absorb one trillion dollars in unrecoverable debt. Every future proposal for a joint crewed Mars mission will be met with one sentence: 'Remember Oblivion.' This clause does NOT apply to Oblivion I-T.

You do not build one ARES-1 — you build three. You do not test the NTP engine until it works — you test it until operational hours exceed mission duration several times over. You do not simulate life support — you run human crews inside it for 600 days on the ground. You do not pre-position one LMO depot — you pre-position two. Schedule is permanently subordinate to certainty.

Part Ten — Summary of Key Facts

Program	The Horizon Program
Test Flight	Oblivion I-T — Uncrewed — \$450 billion
Crewed Mission	Oblivion I — 8 crew — \$550 billion
Total Budget	\$1,000,000,000,000 (flexible by unanimous coalition agreement)
Rocket	Oblivion XXX (XXX = final engine count, TBD)
Spacecraft	ARES-1 — 150m x 15m — vertical integrated — NTP from orbit only
Mission Type	Crewed Mars Orbital Insertion — first humans to orbit Mars
Refueling	Two-point: Earth staging orbit + Low Mars Orbit depot (x2)
Delta-V	~5,900-6,200 m/s total mission
Assembly	7 modules — orbital assembly — 15 m fairing required
Coalition	USA / China / Russia / Saudi Arabia / Japan / India / UAE / Canada
Failure Clause	Crewed mission failure = zero funding, dissolution, full international debt
Design Finalized	April 22, 2026 ARES-1 vertical integrated design locked

One Trillion Dollars. Eight Crew Members. One Mission. No Second Chances.
The Horizon Is Waiting.